

DMRT1 is a testis determining gene in rabbits and is also essential for female fertility

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Abstract

DMRT1 is the testis-determining factor in several species of vertebrates, but its involvement in mammalian testes differentiation, where *SRY* is the testis-determining gene, remains ambiguous. So far, DMRT1 loss of function has been described in two mammalian species and induces different phenotypes: disorders of sex development (46, XY DSD) in men and male infertility in mice. We thus abolished DMRT1 expression by CRISPR/Cas9 in a third species of mammal, the rabbit. First, we observed that gonads from XY *DMRT1*^{-/-} rabbit fetuses differentiated like ovaries, highlighting that DMRT1 is involved in testis determination. In addition to *SRY*, DMRT1 is required in the supporting cells to increase the expression of the *SOX9* gene, which heads the testicular genetic cascade. Second, we highlighted another function of DMRT1 in the germline since XX and XY *DMRT1*^{-/-} ovaries did not undergo meiosis and folliculogenesis. XX *DMRT1*^{-/-} adult females were sterile, showing that DMRT1 is also crucial for female fertility. To conclude, these phenotypes indicate an evolutionary continuum between non-mammalian vertebrates such as birds and non-rodent mammals. Furthermore, our data support the potential involvement of *DMRT1* mutations in different human pathologies, such as 46, XY DSD as well as male and female infertility.

eLife assessment

In this **important** study, the rabbit was used as a non-rodent mammalian model to show that DMRT1 has a testicular promoting function as it does in humans. The experiments are meticulous and **compelling**, and the arguments are clear and **convincing**. These results may explain the gonadal dysgenesis associated with mutations in human DMRT1 and highlight the need for mammalian models other than mice to better understand the process of gonadal sex determination in humans.

DMRT1 (Doublesex and Mab-3 Related Transcription factor 1) belongs to the highly conserved family of DM domain proteins, which exhibits a zinc finger DNA-binding motif that was initially identified in *Drosophila* and *Caenorhabditis elegans* (Erdman and Burtis, 1993; Raymond et al., 1998). Some of its orthologs have been described as Testis Determining Factor (TDF) in vertebrate species such as medaka (*Oryzias latipes*) (Matsuda et al., 2002), xenopus (*Xenopus laevis*) (Yoshimoto et al., 2010), or chicken (Smith et al., 2009). In the last, the Z chromosome carries the *DMRT1* gene. In ZZ males, two copies of the *DMRT1* gene are required to induce testis determination. In ZW females and ZZ chickens harboring a non-functional copy, gonads differentiate as ovaries showing that sex determination is based on DMRT1 dosage (Ioannidis et al., 2021).

In mammals, where the sex-determining system is XX/XY, the TDF is the *SRY* gene (Sex determining Region of the Y chromosome) carried by the Y chromosome. Based on the mouse species, DMRT1 does not appear to have retained a crucial function in testis determination since targeted deletion of *Dmrt1* only affects post-natal testis function. In fact, DMRT1 has roles in both germ cells and supporting cells in the testis, and *Dmrt1*^{-/-} males showed spermatogenesis failure with spermatogonia that did not undergo meiosis (Matson et al., 2010). However, specific knock-out of *Dmrt1* in adult Sertoli cells led to their transdifferentiation into granulosa cells (Matson et al., 2011). Although DMRT1 is not required for testis determination in mice, it retained part of its function in adulthood when it is necessary to maintain Sertoli cell identity. In ovarian differentiation, FOXL2 (Forkhead family box L2) showed a similar function discrepancy between mice and goats as DMRT1 in the testis pathway. In the mouse, *Foxl2* is expressed in female supporting cells early in development but does not appear necessary for fetal ovary differentiation (Uda et al., 2004). On the contrary, it is required in adult granulosa cells to maintain female-supporting cell identity (Ottolenghi et al., 2005; Uhlenhaut et al., 2009). In other mammalian species, such as goats, *FOXL2* was shown to be crucial for ovarian determination. Indeed, naturally observed in the PIS (Polled Intersex Syndrome) mutation (Pailhoux et al., 2001) or experimentally induced by genome editing in goats (Boulanger et al., 2014), *FOXL2* loss-of-function led to female-to-male sex reversal with the early development of XX testes. Following *FOXL2* absence of expression in the XX mutant gonads (XX PIS^{-/-} or XX *FOXL2*^{-/-}), *DMRT1* was up-regulated within days before increased *SOX9* expression, which then directs the differentiation of Sertoli cells and the formation of testicular cords (Elzaïat et al., 2014). These observations in the goat suggested that DMRT1 could retain function in *SOX9* activation and, thus, in testis determination in several mammals. In humans, a few mutations affecting *DMRT1* have been described in patients presenting 46, XY DSD (Disorders of Sex Development) (Chauhan et al., 2017; Ledig et al., 2012; Mello et al., 2010). In particular, a heterozygous *de novo* point mutation in the *DMRT1* gene has been identified in a 46, XY individual with complete gonadal dysgenesis (Murphy et al., 2015), suggesting that DMRT1 and *SRY* may be involved in testicular determination.

To clarify DMRT1 functions in non-rodent mammals, we have chosen the rabbit model, where we generated a DMRT1 mutant line thanks to the CRISPR/Cas9 technology. Firstly, we characterized the DMRT1 expression in control gonads, showing that both XY and XX fetal gonads were expressing *DMRT1* before their sexual differentiation. In XY fetuses, *DMRT1* and *SRY* presented partially overlapping territory, and somatic cells expressing both of them harbored *SOX9* expression and differentiated into Sertoli cells. Secondly, thanks to our CRISPR/Cas9 genetically modified rabbit model, we demonstrated that DMRT1 was required for testis differentiation since XY *DMRT1*^{-/-} rabbits showed early male-to-female sex reversal with differentiating ovaries and complete female genitalia. However, germ cells failed to undergo meiosis, and follicles did not form in XY and XX *DMRT1*^{-/-} mutant ovaries, leading to female infertility. Finally, we demonstrated that DMRT1 was a testis-determining factor in mammals and that it was also required for female fertility.

***DMRT1* is expressed in genital crests of both sexes and just after *SRY* in XY developing testes**

DMRT1 expression pattern has already been reported by molecular analysis in the rabbit species from 14 dpc to adulthood (Daniel-Carlier et al., 2013 [\[1\]](#)). We aimed to investigate further the location of the *DMRT1* expression during gonadal development, firstly at earlier stages of genital crest formation (12–13 days *post-coïtum*, dpc; **Figure 1A** [\[1\]](#)) using *in situ* hybridization (ISH). *SRY* expression was already detected at 12 dpc and, as expected, was found only in the XY genital ridges, where it was restricted to the medullary part of the gonad (**Figure 1B** [\[1\]](#)). By contrast, *DMRT1* was faintly expressed in the gonads of both sexes, in a few cells of the *medulla* under the coelomic epithelium (**Figure 1B** [\[1\]](#)). At 12 dpc, only very few germ cells, expressing *POUSF1*, have completed their migration into the genital ridges (**Figure 1B** [\[1\]](#)). Twenty-four hours later, at 13 dpc, the genital ridges had tripled in size in both sexes, and the territory of *SRY* expression increased within the XY developing testes (**Figure 1C** [\[1\]](#)). The number of somatic cells expressing *DMRT1* was also strongly increased in both sexes, with few of them located in the coelomic epithelium (**Figure 1C** [\[1\]](#)). In addition, more *POUSF1*-expressing germ cells were detected (5 to 12 per section instead of 1 or 2 at 12 dpc) (**Figure 1B** [\[1\]](#) and **1C** [\[1\]](#)).

SOX9 is detected in XY medullar cells co-expressing *SRY* and *DMRT1*

At 14 dpc, the SOX9 protein was immunodetected in a few cells located in the medullary part of the XY gonad (**Figure 2** [\[1\]](#)). Numerous somatic cells of this region also expressed *SRY* and *DMRT1* (**Figure 2** [\[1\]](#)), and a few co-expressed SOX9 and *DMRT1* simultaneously (**Figure 2-figure supplement 1**). By contrast, coelomic epithelial cells only expressed *DMRT1* (**Figure 2** [\[1\]](#)).

At 15 dpc, Sertoli cells that co-express *SRY*, *DMRT1*, and SOX9 began to be organized into embryonic cords (**Figure 2** [\[1\]](#)). At this stage, coelomic epithelial cells expressed *DMRT1* but were negative for *SRY* and SOX9. Furthermore, we observed an islet of cells expressing *SRY* and *DMRT1* located in the mesonephros below the boundary with the gonad (**Figure 2** [\[1\]](#), dotted line). These cells expressed PAX8 (**Figure 2-figure supplement 2**) and could correspond to the recently described supporting-like cell population contributing to the *rete testis* in mice (Mayère et al., 2022 [\[2\]](#)). As in mice, these cells will express SOX9 at the latter stages (a few of them are already SOX9 positive at 15 dpc), but unlike mice, they express *SRY*.

From 16 to 18 dpc, the development of the testicular cords proceeded. At these two stages (16 and 18 dpc), *SRY*, *DMRT1*, and SOX9 were expressed only in the Sertoli cells, where *SRY* expression began to decrease from 18 dpc (**Figure 2** [\[1\]](#)). No more *DMRT1* expression could be seen in the coelomic epithelial cells, but the tunica albuginea begins to form (**Figure 2** [\[1\]](#)), and consequently, the coelomic epithelium will become the surface epithelium.

Persistent expression of *DMRT1* in XX gonadal somatic cells until ovigerous nest formation

As described above, *DMRT1* expression started at 12 dpc in the gonadal somatic compartment of both sexes (**Figure 1B** [\[1\]](#)). In the female gonads, *DMRT1* remained expressed in all somatic cells, including those of the coelomic epithelium, until 16 dpc (**Figure 3A** [\[1\]](#)). Interestingly, as in XY gonads, we observed PAX8-positives cells in XX gonads at 15 dpc (**Figure 3-figure supplement 2**). These cells could contribute to the formation of the *rete ovarii* as in mice (Mayère et al., 2022 [\[2\]](#)). At 18 dpc, *DMRT1* expression decreased but persisted in some cells located in the coelomic epithelium and just below it, where ovigerous nest formation occurred. Interestingly, the female *DMRT1*-antagonist gene *FOXL2* began to be expressed between 16 and 18 dpc when *DMRT1*

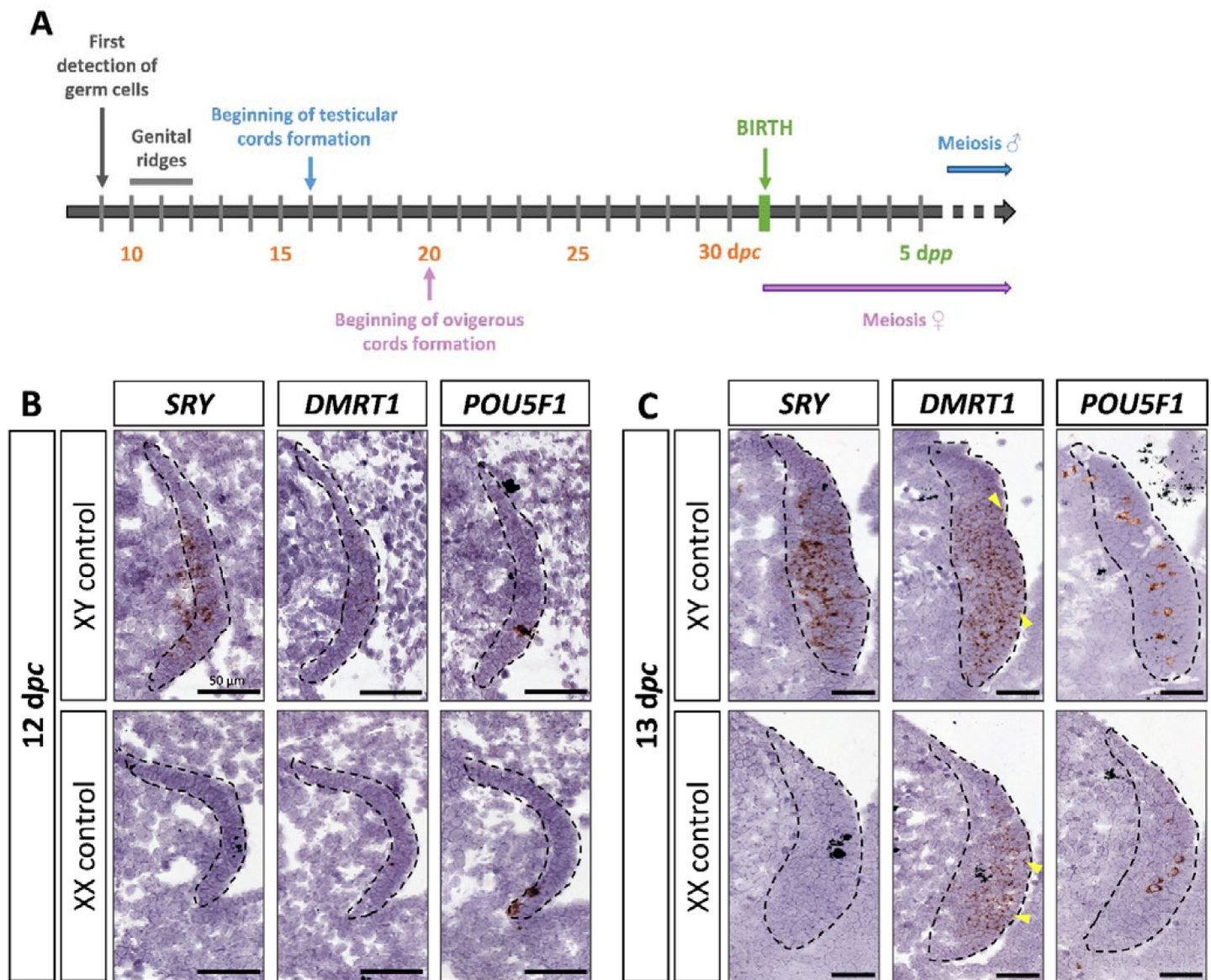


Figure 1.

***SRY*, *DMRT1*, and *POU5F1* location during early gonadal development.**

(A) Key stages of gonadal development in rabbits with 31 days of gestation. Germ cells are first detected at 9 days *post-coitum* (dpc), before the genital ridge formation, which occurs between 10 and 12 dpc. In XY gonads, testicular cords begin forming at 16 dpc, and germ cells enter meiosis a few months after birth. In XX gonads, the ovigerous cords appear at 20 dpc, and meiosis begins around birth. Location of *SRY*, *DMRT1*, and *POU5F1* by *in situ* hybridization (RNAscope technology) on XY and XX control gonads at (B) 12 dpc or (C) 13 dpc. Dotted line: developing genital crests. Yellow arrowheads: coelomic epithelial cells expressing *DMRT1*. Scale bar = 50 μ m.

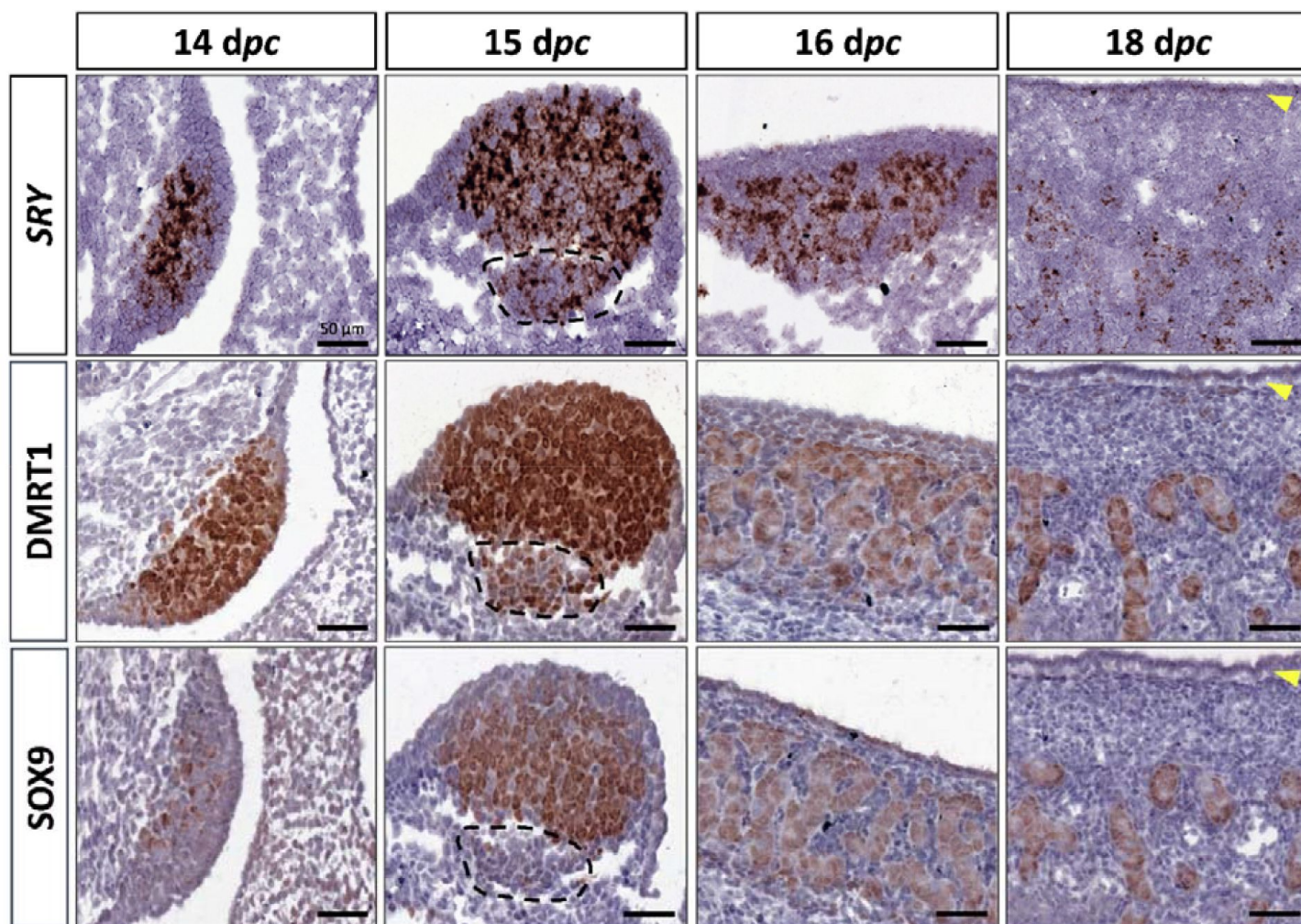


Figure 2.

Somatic markers location during testis differentiation.

Location of *SRY* by *in situ* hybridization (RNAscope technology), *DMRT1*, and *SOX9* by immunohistochemistry on XY control testes from 14 to 18 dpc. The dotted line at 15 dpc: territory with cells expressing *SRY* and *DMRT1* but not *SOX9*. Yellow arrowheads: tunica albuginea in formation. Scale bar = 50 μ m.

expression decreased but persisted in some cells located in the coelomic epithelium and just below it, where ovigerous nest formation occurred. Interestingly, the female *DMRT1*-antagonist gene *FOXL2* began to be expressed between 16 and 18 dpc when *DMRT1* expression decreased (**Figure 3B**). Thereafter, at 20 dpc, *DMRT1* expression was limited in some somatic cells enclosed in nascent ovigerous nests where some germinal cells also began to be positive for *DMRT1* (**Figure 3C** and **Figure 3-figure supplement 3**). At this stage, *DMRT1* positive territory seems to overlap that of *RSPO1* but not that of *FOXL2* located in the loose conjunctive tissue around the ovigerous nests (**Figure 3C**).

The testicular formation is impaired in *DMRT1* knock-out XY rabbits

To determine the role of *DMRT1* in the rabbit species used as a non-rodent mammalian model, we engineered a *DMRT1* knock-out line using the CRISPR/Cas9 system with two RNA guides located in exon 3. The mutation carried by this line is a 47-bp duplication in sense, leading to a frameshift of the open reading frame and a premature stop codon (**Figure 4-figure supplement 4A**). This mutation does not affect *DMRT1* transcription but induces a total absence of protein as shown in post-natal gonads by western blot (**Figure 4-figure supplement 4B and C**). Thanks to this line, we first analyzed gonadal formation at 20 dpc, when the testis and ovary were distinguishable in control animals. Indeed, at this stage, testes appeared with well-formed seminiferous cords, and ovigerous nest formation was clearly in progress in the ovaries (**Figure 4A**). At 20 dpc, XY *DMRT1*^{-/-} gonads failed to engage testicular differentiation and appeared quite like control ovaries, but ovarian differentiation did not appear to be affected by the loss of *DMRT1* (**Figure 4A**). To better characterize the *DMRT1*^{-/-} gonads in XY and XX fetuses, we established the gonadal transcriptome by RNA-sequencing. Heatmap representation of the 3640 differentially expressed genes in at least one of the four genotypes (adjusted p-value <0.05 and |log2FC|>1; **Figure 4-table supplement 1**) was clustered into eight groups (#1 to #8, **Figure 4B** and **Figure 4-table supplement 2**). Clusters #1 and #7 contained 1331 and 315 genes respectively, which were preferentially expressed in XY control testes. Expression of these genes was decreased in XY *DMRT1*^{-/-} gonads, harboring levels close to that of the female's ovaries (XX control or *DMRT1*^{-/-}). On the other hand, clusters #2, #3, and #5 (537, 582, and 464 genes respectively) were composed of genes preferentially expressed in XX control ovaries, and their expression was increased in XY *DMRT1*^{-/-} gonads. Deep-sequencing transcriptomics confirmed the ovarian fate of XY *DMRT1*^{-/-} gonads. The heatmap in **Figure 4C** also illustrates the expression for selecting some of the main genes involved in sex determination (**Figure 4C**).

Expression levels and patterns of the principal actors of gonadal differentiation were confirmed by RT-qPCR, and the location of positive cells was achieved by immunohistochemistry. As expected, *SOX9*, *AMH*, and *DHH* expression levels were decreased in XY *DMRT1*^{-/-} gonads, remaining like those detected in control or *DMRT1*^{-/-} XX ovaries, while *SRY* expression was enhanced in XY *DMRT1*^{-/-} gonads (**Figure 5A**). Interestingly, we noticed a slight increase of *SOX9*-positive cells in XY *DMRT1*^{-/-} gonads compared to XX control or mutant ovaries (**Figure 5C**). By contrast, *FOXL2* and *CYP19A1* expression were increased in XY *DMRT1*^{-/-} gonads to similar levels to those detected in control or mutant ovaries (**Figure 5B**). By immunohistochemistry, we detected cells expressing *FOXL2* in XY *DMRT1*^{-/-} gonads (**Figure 5C**). Moreover, *RSPO1* expression was increased in XY *DMRT1*^{-/-} gonads, but it remained lower than in control ovaries or in XX *DMRT1*^{-/-} gonads. In the latter, the *RSPO1* expression was also lower than in control ovaries, suggesting a regulatory link between *DMRT1* and *RSPO1* in the female pathway (**Figure 5B**).

Germ cells failed to engage meiosis in *DMRT1* mutant gonads

After the sex determination process and the first stages of gonad formation, *DMRT1*^{-/-} gonads engage a female fate and differentiate as ovaries, whatever their sex-chromosome constitution, XX or XY. Whereas the *DMRT1* expression began at 18 dpc in the XY germinal lineage of control gonads and 20 dpc in XX (**Figure 6-figure supplement 3**), its expression was abolished in both

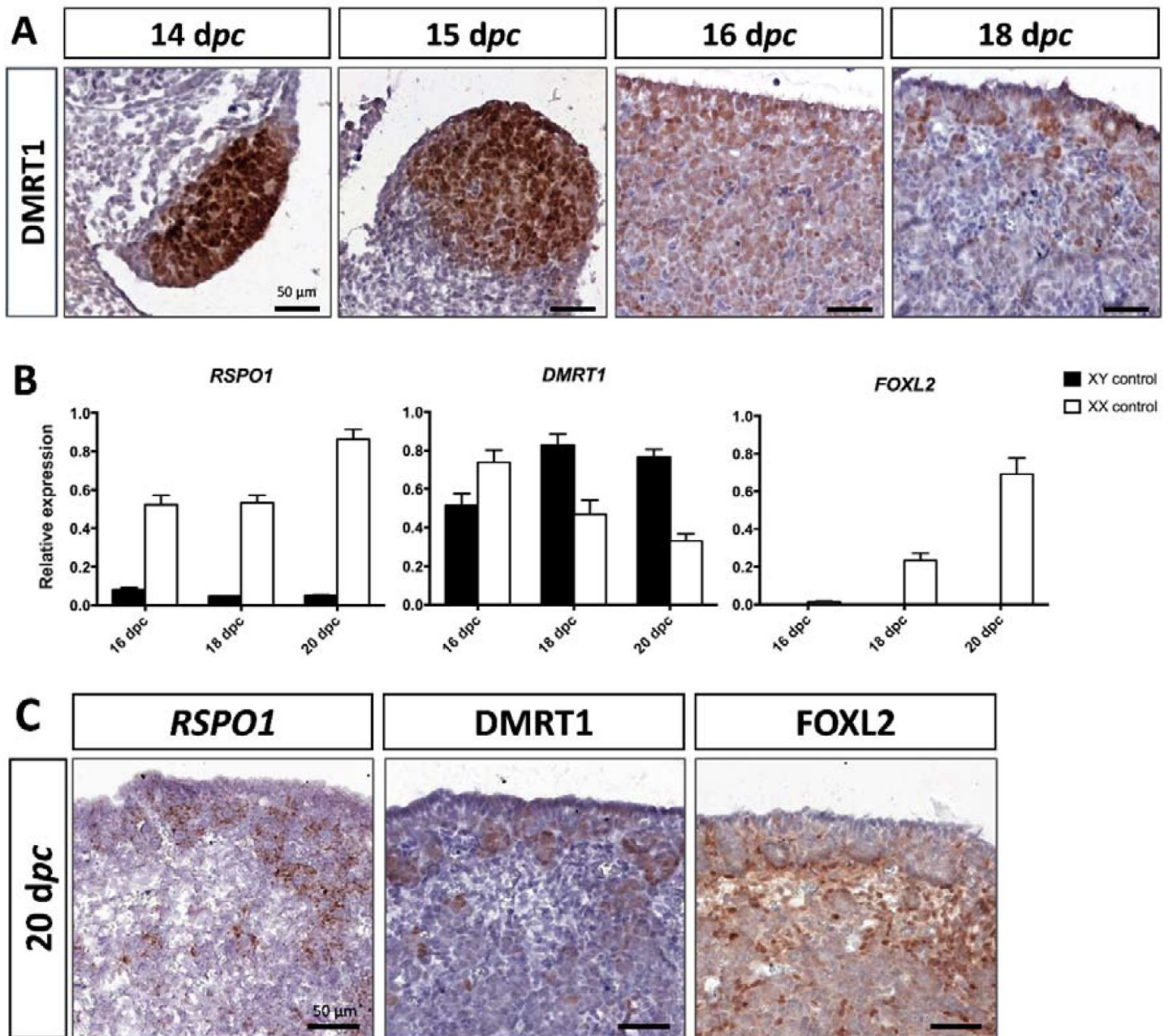


Figure 3.

Somatic markers location and expression during ovarian differentiation.

(A) Immunostaining of DMRT1 on XX control gonads from 14 to 18 dpc. (B) RT-qPCR analyses of *RSPO1*, *DMRT1*, and *FOXL2* expression from 16 to 20 dpc in control gonads of both sexes (n=3-5). (C) *RSPO1* *in situ* hybridization (RNAscope technology), immunostaining of DMRT1 and FOXL2 on 20 dpc control ovaries. Scale bar = 50 μ m.

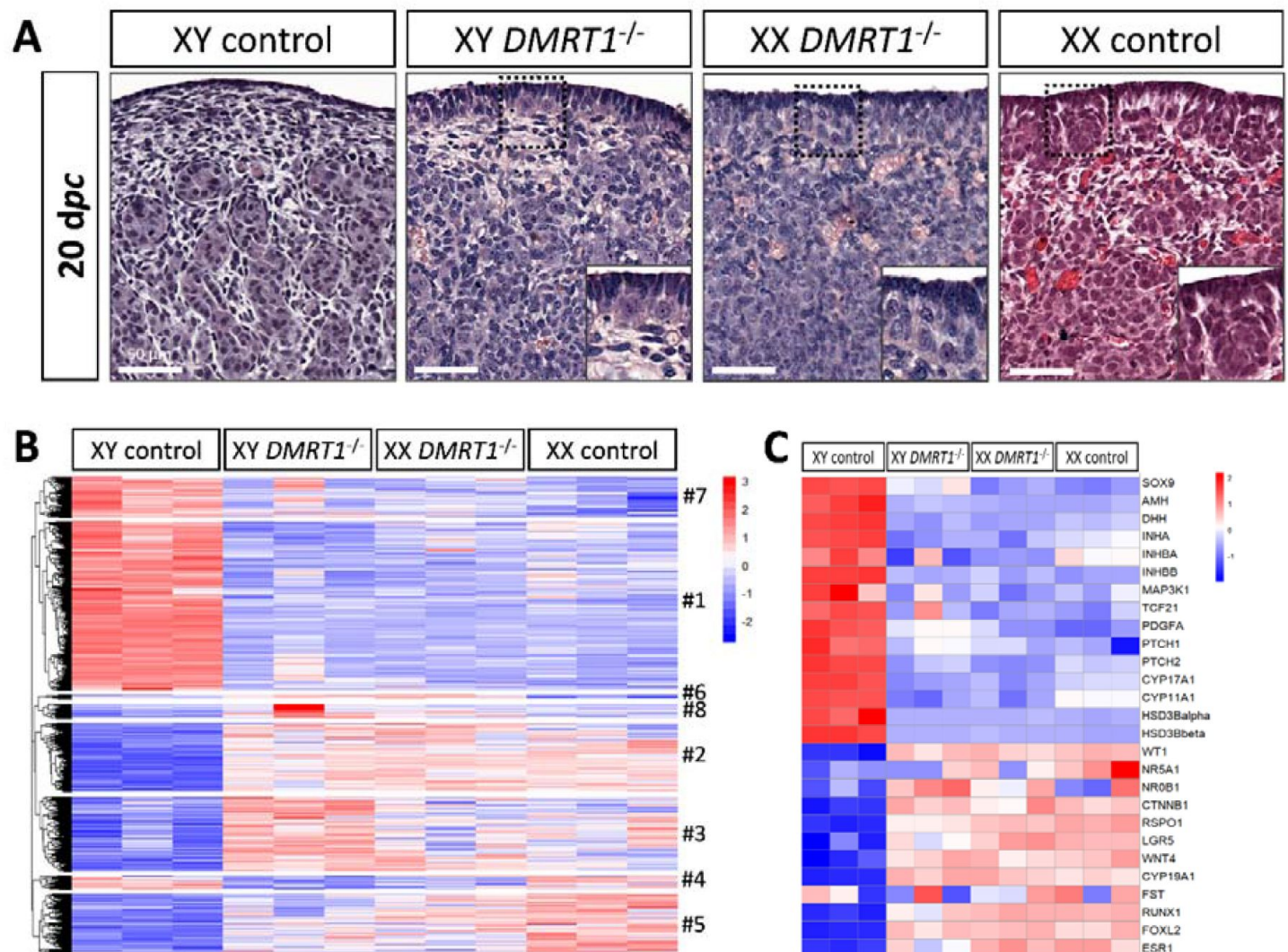


Figure 4.

Ovarian-like morphology and transcriptomic signature of XY *DMRT1*^{-/-} gonads at 20 dpc.

(A) Hematoxylin and eosin staining of gonads sections from control and *DMRT1*^{-/-} 20 dpc rabbits. The enlarged area shows the characteristic ovarian surface epithelium found on XY *DMRT1*^{-/-} gonads. Scale bar = 50 μ m. Heatmap representation of (B) 3460 deregulated genes (adjusted *p*-value <0.05 and |log2FC|>1) or (C) 27 selected genes between XY control, XY *DMRT1*^{-/-}, XX *DMRT1*^{-/-}, and XX control at 20 dpc.

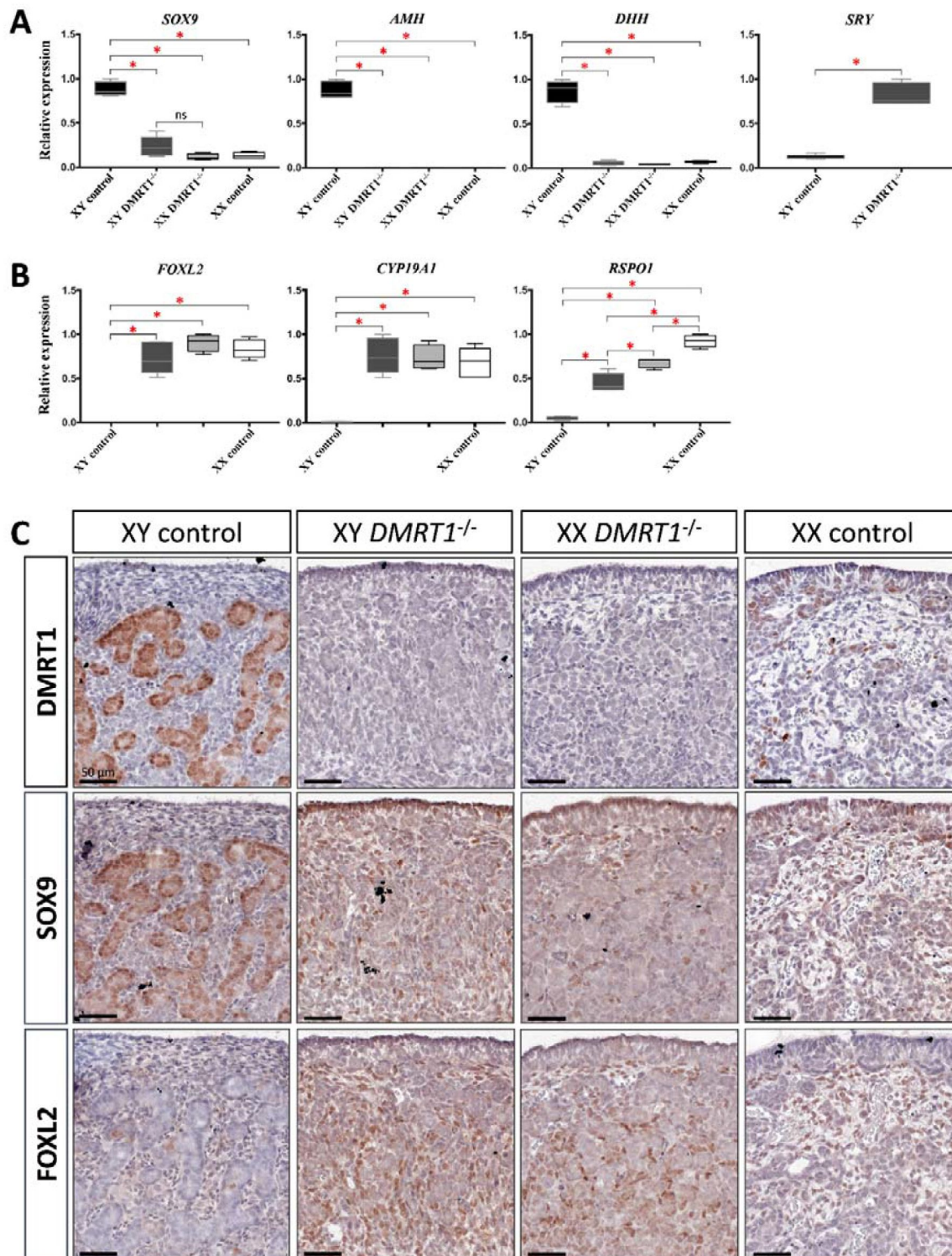


Figure 5.

Somatic markers expression and location on control and *DMRT1*^{-/-} gonads at 20 dpc.

RT-qPCR analyses of (A) testicular-related differentiation genes (*SOX9*, *AMH*, *DHH*, and *SRY*) or (B) ovarian-related differentiation genes (*FOXL2*, *CYP19A1*, and *RSPO1*) in XY control, XY *DMRT1*^{-/-}, XX *DMRT1*^{-/-} and XX control gonads (n=4-5) at 20 dpc. Statistical analyses were performed using the non-parametric Kruskal-Wallis test, followed by a pairwise permutation test: * *p*-value < 0.05; ns: non-significant. (C) Immunostaining of *DMRT1*, *SOX9*, and *FOXL2* on XY control, XY *DMRT1*^{-/-}, XX *DMRT1*^{-/-}, and XX control gonad sections at 20 dpc. Scale bar = 50 μ m.

somatic and germ cells in *DMRT1*^{-/-} mutant gonads (**Figure 5C**). Although XX or XY *DMRT1*^{-/-} gonads continue to develop as ovaries, most germ cells did not engage in the meiotic process. Indeed, in control ovaries at 3 days *post-partum* (dpp), most germ cells were in the zygotene stage, showing nuclei with highly condensed chromatin (Daniel-Carlier et al., 2013) (**Figure 6**) and were positives for Ki67, showing their exit from the G0 phase of the cell cycle (**Figure 6-figure supplement 5**). By contrast, in *DMRT1*^{-/-} gonads, few germ cells in the preleptotene stage were observed (**Figure 6**), and the majority did not express Ki67 but continued to express the pluripotency marker POU5F1 (**Figure 6-figure supplement 5**). Subsequently, the rupture of ovarian nests and the follicle formation did not occur in *DMRT1*^{-/-} gonads. At 18 dpp, folliculogenesis had already started in control ovaries, where the first primordial follicles were visible in the deepest cortical part close to the *medulla* (**Figure 6**). By contrast, *DMRT1*^{-/-} gonads seemed to be blocked at a pre-meiotic stage, and folliculogenesis failed to occur (**Figure 6**). In adults, *DMRT1*^{-/-} gonads were reduced in size (**Figure 6-figure supplement 6**), no germ cells were detected, and some somatic cells evolved toward luteinized cells (**Figure 6**). Consequently, both XY and XX females were completely infertile in adulthood.

Discussion

Our study gave new insights into the conservation of the sex-determination genetic cascade across evolution. Although the signal controlling this process could take different forms in metazoans, several downstream transcription factors involved in gonadal differentiation have been conserved throughout evolution. For instance, *SOX9*, well known in vertebrates as being essential for Sertoli cell differentiation (Chaboissier et al., 2004; Foster et al., 1994; Qin and Bishop, 2005; Vidal et al., 2001; Wagner et al., 1994), has a fruit fly ancestor, *Sox100B*, which was found to be necessary for testis development in *Drosophila* (Nanda et al., 2009). However, the most conserved sex-differentiating factor throughout evolution is *DMRT1*. Indeed, it has been maintained at the head of the sex determination cascade in reptiles (Sun et al., 2017), fishes (Matsuda et al., 2002), and birds (Smith et al., 2009). Nevertheless, its functions could have been reduced in mammals since testis differentiates in the absence of *DMRT1* (*Dmrt1*^{-/-}) in mice (Raymond et al., 2000). Our results highlight an evolutionary continuum of this gene in testis determination from birds to rabbits and non-rodent mammals in general. Interestingly, even *DMRT1* dosage sensibility has been conserved between chicken and rabbits since heterozygous XY *DMRT1*^{+/-} male rabbits present secondary infertility with an arrest of spermatogenesis around two years of age (data not shown).

DMRT1 position in the rabbit sex-determining cascade

As the early stages of gonadal differentiation in rabbits were not fully characterized, we first determined the expressional profiles of the major sex-determining genes. We observed that *DMRT1* expression started at 12 dpc, at the early formation of genital crests, and it was first expressed in the somatic lineage of both sexes, as in mice (Lei et al., 2007; Raymond et al., 1999) or in humans (Garcia-Alonso et al., 2022). In the human fetal testis, *DMRT1* expression is co-detected with *SRY* in early supporting gonadal cells (ESCGs), which become Sertoli cells following the activation of *SOX9* expression (Garcia-Alonso et al., 2022). In mice, the *Dmrt1* expression starts at E10.5 in both somatic and germinal compartments. However, we showed that germline expression was shifted by 6 to 8 days compared to the somatic compartment in the rabbit male and female gonads, respectively. These differences are strongly related to the timing of gonadal development in rabbits - which is longer than in mice - and therefore allows better visualization of the different processes. These sequential *DMRT1* up-regulations according to cell type and sex also argue in favor of distinct *DMRT1* promoters as already described in rats (Lei et al., 2009). For the somatic XY compartment, *SOX9* expression appears at 14 dpc in cells expressing both *DMRT1* and *SRY*, suggesting that both factors are required for *SOX9* up-regulation. This led to the Sertoli cell differentiation and testicular cords formation from 15 dpc. In the

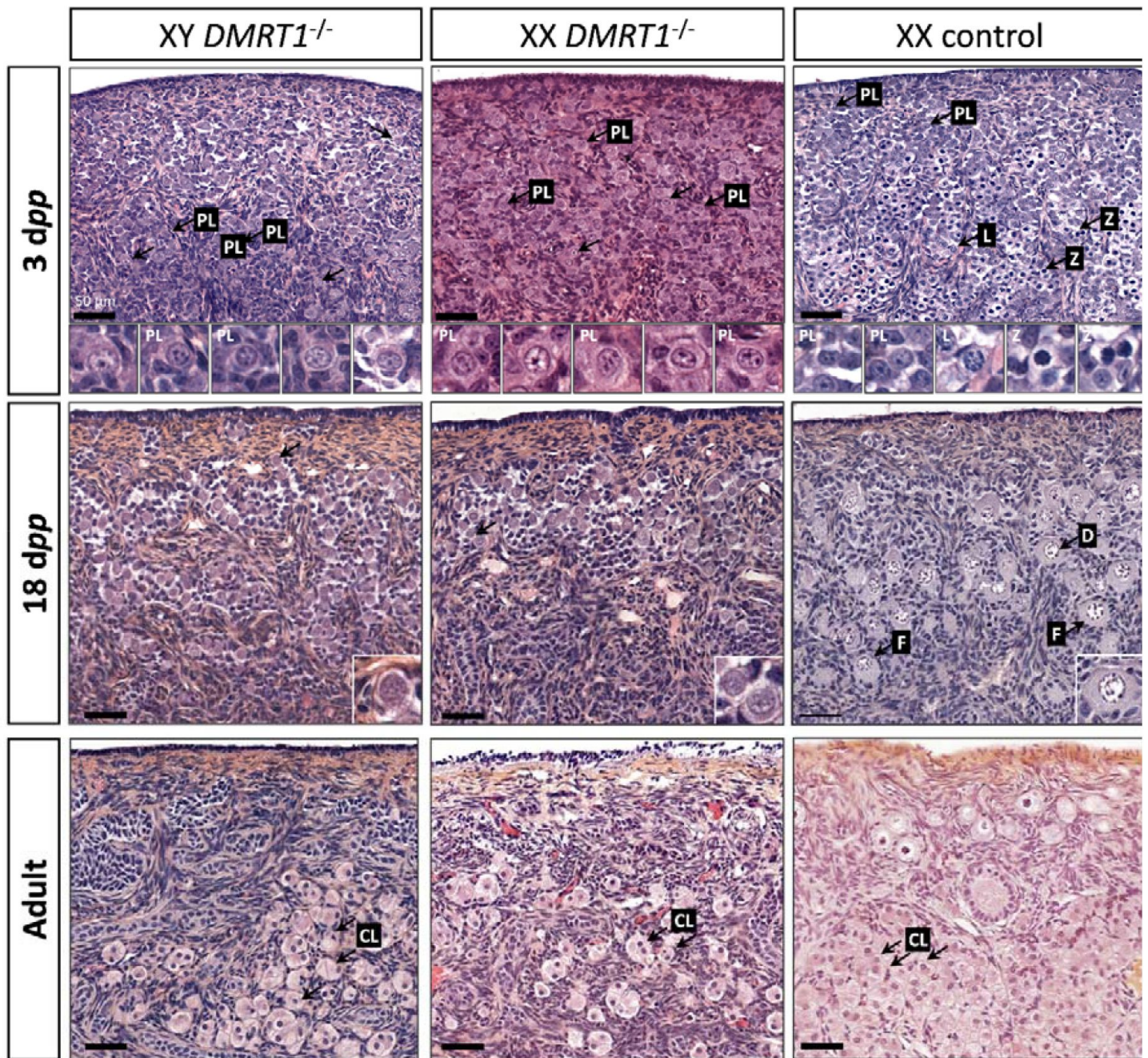


Figure 6.

Evolution of gonadal morphogenesis in XY and XX *DMRT1*^{-/-} rabbits.

Hematoxylin and eosin staining of gonad sections from XY and XX *DMRT1*^{-/-} gonads and XX control ovaries at 3 days *post-partum* (dpp), 18 dpp, and in adulthood (4-9 months). The enlargements for the first two panels correspond to the nuclei pointed by an arrow. PL: preleptotene stage; L: leptotene stage; Z: zygotene stage; D: diplotene stage; F: ovarian follicle; CL: luteal cells. Scale bar = 50 µm.

developing ovary, we showed that *FOXL2* increases when *DMRT1* expression starts to shift from somatic cells to germ cells. Moreover, our results suggested *DMRT1* involvement in *RSPO1* up-regulation in the ovary.

DMRT1 is required for testis determination in rabbits

In recent years, the advent of new genome editing technologies has made it possible to explore other animal models, such as the goat (Boulanger et al., 2014) or the rabbit (Jolivet et al., 2022), and enriching our knowledge on the conservation of ancestral genetic mechanisms in non-rodent mammals. In rabbits, the CRISPR-Cas9 technology allowed us to generate a null mutation of the *DMRT1* gene, leading to an absence of detectable protein at homozygosity. Thanks to this model, we could demonstrate that *DMRT1* kept its leadership in sex determination also in mammals, where *SRY* stays the “switch-on factor” for testis determination, as previously demonstrated in rabbits (Song et al., 2017). Very early in fetal life, XY fetuses expressing *SRY* but lacking *DMRT1* (*DMRT1*^{-/-}) presented a male-to-female sex reversal. Although *SRY* expression was maintained in XY homozygous mutant gonads, the activation of *SOX9* expression was weak in the absence of *DMRT1*. Accordingly, a few cells expressing *SOX9* protein were detectable, but *SOX9* target genes expression were not activated in XY *DMRT1*^{-/-} gonads. Thus, *DMRT1* seems to be required for *SRY* action on its targets (i.e., *SOX9* gene activation) but also for *SOX9* functions in the early fetal gonad. Interestingly, a recent study proposed that *DMRT1* can act as a *SOX9* pioneer factor in the post-natal testis for Sertoli cell identity maintenance (Lindeman et al., 2021). In rabbits, *DMRT1* is required for *SOX9* and *SRY* functions, and we hypothesize that *DMRT1* might be a pioneer factor for both. In the differentiating genital crest, *DMRT1* would be required to increase chromatin accessibility on specific sex-related regions, allowing *SRY* to bind and activate its targets and particularly the expression of *SOX9*. The crucial region for *SRY* binding was identified in mice more than 500 kb upstream of the *Sox9* transcription start site and named Enhancer 13 (Gonen et al., 2018). Conservation studies identified the homolog of Enhancer 13 in many mammalian species, including humans, cows, and rabbits, and *DMRT1* consensus sites were predicted in all mammals examined except mice and rats (Gonen et al., 2018). In non-rodent mammals, *DMRT1* might be required for chromatin remodeling on the Enhancer 13 region to enable *SRY* binding and *SOX9* expression since the beginning of testis differentiation. In the mouse, which evolved more rapidly, *DMRT1* would no longer be necessary for *SRY* action because the chromatin state of the fetal supporting cells would be more permissive. This could also explain why *DMRT1* does not exert any critical function in the fetal testis in mice (Raymond et al., 2000). By contrast, it is required for the action of *SOX9* in the post-natal testis (Lindeman et al., 2021), where a sex-specific epigenetic signature was observed (Garcia-Moreno et al., 2019).

DMRT1 is required for germ cell meiosis and female fertility

In addition to its functions in testis differentiation, *DMRT1* also plays a crucial role in the female gonad. Indeed, germ cells did not undergo meiosis in *DMRT1*^{-/-} ovaries, and in the absence of oocyte I, germ cell cysts do not break, compromising follicle formation and female fertility. This specific phenotype is highly similar to those observed in ZW chicken ovaries lacking *DMRT1* (Ioannidis et al., 2021), but is quite different from those described in mice. Even though fewer follicles were observed in *Dmrt1*^{-/-} mice ovaries, the female remains fertile (Krentz et al., 2011). Interestingly in humans, one case involving *DMRT1* in premature ovarian failure has been reported (Bartels et al., 2013).

In rabbit fetuses, *DMRT1* expression was first detected in differentiating ovarian somatic cells, at least until *FOXL2* up-regulation. However, *DMRT1* has also been observed in fetal germ cells from 20 dpc until meiosis proceeded after birth. Consequently, germ cell pre-meiotic arrest in *DMRT1*^{-/-} XX gonads could result from *DMRT1* loss-of-function in the germinal or the somatic compartment or both. In the somatic compartment, the absence of *DMRT1* in XX homozygous mutants did not seem to disturb the first steps of ovarian differentiation. Nevertheless, deep sequencing transcriptomics revealed the dysregulated expression of a few genes involved in the WNT/beta-

catenin pathway. In particular, *RSPO1*, a positive regulator of the WNT signaling, was reduced, and *DKK1*, a negative regulator, was increased (*Table supplement 1* and *table supplement 2*). These two events could have the effect of limiting the beta-catenin action in both somatic and germinal ovarian cells at the beginning of their differentiation. This pathway has proven to be crucial in mice to promote germ cell meiosis (Le Rolle et al., 2021 [↗](#)). Nevertheless, it cannot be the main event explaining the pre-meiotic failure, and the functions of DMRT1 in germ cells are more certainly involved. In mice, DMRT1 was shown to be involved in *Stra8* up-regulation in female germ cells and was thus related to the meiotic process (Krentz et al., 2011 [↗](#)). This regulatory action also seems to be done in close collaboration with the retinoic acid pathway (Feng et al., 2021 [↗](#)). The sole action of DMRT1 on *STRA8* up-regulation cannot explain the phenotype observed in rabbits where the germ cell seems to be unable to leave their pluripotency stage. It has also been demonstrated in male mice that DMRT1 acts as a regulator of spermatogonia pluripotency by directly regulating different pluripotency-associated genes, including *POU5F1* (Krentz et al., 2009 [↗](#); Zhang et al., 2016 [↗](#)). This path is under exploration in our model in order to try to decipher further the critical role of DMRT1 in the germ line of both sexes.

Materials and Methods

Animals

New Zealand rabbits (NZ1777, Hypharm, Roussay, France) were bred at the SAAJ rabbit facility (Jouy-en-Josas, France). All experiments were performed with the approval of the French Ministry MENESR (accreditation number APAFIS#685 and #21451) and following the guidelines issued by the local committee for ethics in animal experimentation (COMETHEA, Jouy-en-Josas). All scientists working directly with the animals possessed an animal experimentation license delivered by the French veterinary services. Hormonal superovulation treatments and surgical embryo transfer procedures were performed as previously described (Peyny et al., 2020 [↗](#)).

Generation of mutant rabbits

Two guide RNAs were designed (<http://crispor.trefor.net/> [↗](#)) to target the third exon, as shown in figure supplement 4A. Embryos produced from superovulated females were injected at the single-cell stage with a mixture of the two sgRNAs (10 ng/μl each) and the Cas9mRNA (10 ng/μl) in the injection buffer. Injected embryos were implanted 3-4 hours after into the oviducts of anesthetized recipient rabbits via laparotomy. Details concerning the handling of females and embryos have been described elsewhere (Peyny et al., 2020 [↗](#)).

Offspring were screened for the presence of InDel mutations using genomic DNA extracted from ear clips (Jolivet et al., 2014 [↗](#)). Founders were detected by PCR using one set of primers (*table supplement 3*) surrounding the position of the targeted region in exon III (*figure supplement 4*). The amplified fragment was sequenced (Eurofins Genomics, Courtaboeuf, France), and the mutation was deduced by comparing it with the sequence of a wild-type rabbit. The same set of primers was used for the routine screening of descendants. The presence/absence of the Y chromosome was deduced from the amplification of the *SRY* gene through PCR analyses (*table supplement 3*). In the present paper, mentions of the XY or XX genotype always refer to the PCR determination.

XY and XX DMRT1^{+/-} rabbits were viable until adulthood and did not appear to have any diseases. DMRT1^{-/-} mutants were obtained by crossing XY DMRT1^{+/-} and XX DMRT1^{+/-} animals.

Histological and Immunohistological analyses

Immediately after sampling, whole embryos or gonads were immersed in 4% paraformaldehyde (PFA) in phosphate-buffered saline (PBS) or Bouin's fixative. After 72 to 96 hours of fixation at 4°C, tissues were washed three times with PBS, and stored at 4°C in 70% ethanol until paraffin

inclusions. Adjacent sections of 5 μm thick were processed using a microtome (Leica RM2245) and organized on Superfrost Plus Slides (J18000AMNZ, Epremedia). Before staining or experiments, sections were deparaffinized and rehydrated in successive baths of xylene and ethanol at room temperature.

Hematoxylin-eosin-saffron (HES) staining was performed by the @Bridge platform (INRAE, Jouy-en-Josas, France) using an automatic Varistain Slide Stainer (Thermo Fisher Scientific). *In situ* hybridization (ISH) was performed using the RNAscope ISH methodology (ACD, Bio-Techne SAS, Rennes, France) when no reliable antibody could be used to characterize the target protein. Briefly, 5- μm sections from PFA-fixed tissue were labeled using RNAscope 2.5HD assay-brown kit (322310, ACD) and 1000 nucleotides long probes designed and produced by the manufacturer (list of all synthesized probes used in *table supplement 4*). Brown labeling was observed as a visible signal, and hybridization was considered to be positive when at least one dot was observed in a cell.

Immunohistochemistry (IHC) was performed using the ABC amplification signal kit (PK-6100, Vector Laboratories) and DAB enzymatic reaction (SK-4100, Vector Laboratories). Briefly, the antigenic sites were unmasked with a citrate buffer (pH 6; H-3300, Vector Laboratories), and endogenous peroxidases were blocked with a 3% H_2O_2 solution (H1009, Sigma-Aldrich). Sections were then permeabilized with 1X PBS, 1% Bovine Serum Albumin (A7906, Sigma-Aldrich), and 0.2% Saponin (7395, Merck) and incubated overnight at 4°C with primary antibodies (*table supplement 5*). Following PBS washes, sections were incubated with biotinylated secondary antibodies (*table supplement 5*). After ABC kit incubation and DAB revelation, hematoxylin staining was briefly performed to visualize the whole tissue.

Immunofluorescence (IF) was performed using Tyramide SuperBoost kit for primary rabbit antibody (B40944, ThermoFisher) as recommended by the manufacturer. Other secondary antibodies used are listed in *table supplement 5*.

All stained sections were scanned using a 3DHISTECH panoramic scanner at the @Bridge platform (INRAE, Jouy-en-Josas, France).

Total RNA extraction and Quantitative RT-PCR (RT-qPCR)

Immediately after sampling, 16-20 *dpc* gonads were snap-frozen in liquid nitrogen and stored at -80°C until extraction. Total RNAs were isolated using Trizol reagent (15596018, Life Technologies), purified with the RNeasy Micro kit (74004, Qiagen) following the manufacturer's instructions, and then DNase-treated (1023460, Qiagen). RNAs were quantified with a Qubit[®] Fluorometric Quantification kit (Q32852, Life Technologies).

Reverse transcription of 50-100 ng RNAs using the Maxima First-Strand cDNA Synthesis Kit (K1641, ThermoScientific) was down. qPCR with diluted cDNA was performed in duplicate for all tested genes with the Step One system (Applied Biosystems) and Fast SYBR[™] Green Master Mix (4385612, Applied Biosystems). *H2AFX* and *YWHAZ* or *SF1* (Splicing Factor 1) were used as the reference genes to normalize the results with qBase⁺ software (Biogazelle NV, Ghent, Belgium). The sequences of the primers used are listed in *table supplement 3*.

For each experiment, values were plotted using GraphPad Prism Software (GraphPad Software Inc., La Jolla, CA, USA). Statistical analyses of data from 20 *dpc* control and *DMRT1*^{-/-} gonads were performed under R studio software. Because of the small number of samples in each group, comparisons were made using the Kruskal-Wallis rank sum test followed by pairwise permutation t-tests (1000 permutations, p-value adjusted with the Benjamin-Hochberg method).

Nuclear proteins extraction and Western-Blot

Gonads from newborns (1-3 days *post-partum*) rabbits were collected and snap-frozen in liquid nitrogen and then stored at -80°C . Frozen gonads were crushed in liquid nitrogen using a mortar. Powdered tissue samples were immediately resuspended in homogenization buffer (10mM Hepes pH7.7; 25mM KCl; 2mM Sucrose; 0.5mM EGTA pH8; 0.15mM Spermin; 0.5mM Spermidin; 0.5mM Dithiothreitol (DTT); 2mM Benzamidin; 0.5mM Phenylmethylsulfonyl fluoride (PMSF); cOmplete, Mini, EDTA-free Protease Inhibitor Cocktail (Roche, 118361700001)) plus 0.3% IGEPAL (3021, Sigma-Aldrich). After centrifugation for 15 min at 4°C and 3,500 rpm, supernatants containing cytosolic proteins were stored at -80°C . Pellets were centrifugated for 1h at 4°C and 12,700 rpm and then resuspended with [C-NaCl] buffer (20mM Hepes pH7.7; 1.5mM MgCl_2 ; 0.2mM EDTA; 25% glycerol; 0.5mM PMSF; 0.5mM DTT; 2mM Benzamidin; cOmplete, Mini, EDTA-free Protease Inhibitor Cocktail). NaCl was added, lysates were rotated for 1h at 4°C and then centrifugated for 30 min at 4°C and 12,700 rpm. Supernatants containing nuclear extracts were collected, and the amount of protein was determined by the Bradford method.

Protein (20 μg of each sample) was separated on 4-15% polyacrylamide gel (456-1083, Bio-Rad) and then transferred into a polyvinylidene difluoride (PVDF) membrane. The membrane was blocked in 4% milk (Difco Skim Milk #232100 diluted in PBS-Tween 2%) and incubated overnight with primary antibodies mouse anti-DMRT1 (S5 Table) or mouse anti-beta Actin (1/5000; GTX26276, Genetex). After washes, the membrane was incubated for 1h at room temperature with the secondary antibody anti-mouse IgG peroxidase-conjugated (1/500; A5906, Sigma-Aldrich). The revelation was performed using Pierce™ ECL Plus Western Blotting Substrate (32312, ThermoFisher), and the signal was observed with the Chemi-Doc Touch Imaging System (Bio-Rad). For rehybridization, the membrane was stripped for 10 min in Restore™ Western Blot stripping buffer (21059, ThermoFisher).

RNA-sequencing and bioinformatics analysis

Total RNAs were extracted from control and *DMRT1*^{-/-} rabbit gonads at 20 dpc (n=3 for each phenotype and each sex). Total RNA quality was verified on an Agilent 2100 Bioanalyser (Matriks, Norway), and samples with a RIN > 9 were made available for RNA sequencing. This work benefited from the facilities and expertise of the I2BC High-throughput Sequencing Platform (<https://www.i2bc.paris-saclay.fr/sequencing/ng-sequencing/> Université Paris-Saclay, Gif-sur-Yvette, France) for oriented library preparation (Illumina Truseq RNA Sample Preparation Kit) and sequencing (Paired-end 50-35 bp; NextSeq500). More than 37 million 50-35 bp paired-end reads per sample were generated. Demultiplexing was done (bcl2fastq2-2.18.12), and adapters were removed (Cutadapt1.15) at the I2BC High-throughput Sequencing Platform. Only reads longer than 10 pb were used for analysis. Quality control of raw RNA-Seq data was processed by FastQC v0.11.5.

Reads were mapped on all the genes of a better-annotated rabbit genome. Indeed, we improved the current reference rabbit transcriptome (OryCun2.0; *Oryctolagus cuniculus*, Ensembl version 106). For this purpose, we extended the 5' and 3'-UTRs of genes using rabbit gonad RNA-seq data available in public databases (<https://www.ncbi.nlm.nih.gov/bioproject/PRJEB26840>). In addition, the annotation and for some of them, sequences of 22 marker genes of gonadal differentiation missing or wrong in OryCun2.0 was added or fixed to this genome assembly. Then, after mapping with STAR version 2.5.1b (Dobin et al., 2013), reads were counted using FeatureCounts version 1.4.5 (Liao et al., 2014). Data normalization and single-gene level analyses of differential expression were performed using DESeq2 (Love et al., 2014). Differences were considered to be significant for Benjamini-Hochberg adjusted p-values [<0.05 , and absolute fold $\text{Log}_2\text{FC} > 1$ (Benjamini and Hochberg, 1995)]. RNA-seq data were deposited via the SRA Submission portal (<https://www.ncbi.nlm.nih.gov/sra/PRJNA899447>), BioProject ID PRJNA899447.

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Competing interests

Authors declare that they have no competing interests.

Data and materials availability

The datasets generated for this study can be found in the sequence read archive at: <https://www.ncbi.nlm.nih.gov/sra/PRJNA899447> 

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Reviewer #1 (Public Review):

DMRT1 is essential in testis development in different species. While *Dmrt1* is the testis-determining factor in chicken and deletion encompassing this gene lead to gonadal dysgenesis in human, the role of DMRT1 in testis development remains to be clarified. Despite an early expression of *Dmrt1* in the mouse gonad and a potential function as a pioneer factor, DMRT1 is only required for the maintenance of the Sertoli cell identity in the postnatal testis. The use of a new animal model could provide new insights into the role of this factor in humans. Here the authors have generated a knockout model of DMRT1 in rabbits. They show that the XY mutant gonads differentiate as ovary indicating that DMRT1 is required for testis differentiation in rabbits. In addition, most of the germ cells remain pluripotent as evidenced by the maintenance of *POU5F1* in both XY and XX mutant gonads. These are very important results potentially explaining gonadal dysgenesis associated with the DMRT1 locus in disorders of sex development in humans.

The experiments are meticulous and convincing. I find the arguments of the authors about the role of DMRT1 in germ cells in addition to its function in Sertoli cell differentiation, both comprehensible and compelling. Clearly, this is an important insight in sex determination and gametogenesis.

Reviewer #2 (Public Review):

It is well known that DMRT proteins and more specifically, DMRT1 plays a key role in the sex determination processes of many species. While DMRT1 has been shown to be critical for the sex determination of fish, birds, and reptiles, it seems less crucial at the sex determination stages of the mice. It is important though for adult sex maintenance in mice.

Unlike its minor role in mouse sex determination, it seems that variants in DMRT1 in humans cause 46, XY DSD and sex reversal.

The paper by Dujardin et al. is a beautiful study that provides an answer to this long-lasting discrepancy of the difference between the two common mammal species: human and mouse.

It is a really nice example of how working with other mammal species, like the rabbit, could serve as a nice model for understanding mammalian sex determination.

In this study the researchers first described the expression patterns of DMRT1 in the rabbit XY and XX gonads throughout the window of sex determination.

They then used CRISPR/Cas9 to generate DMRT1 KO rabbits and analysed the phenotype in XY and XX rabbits. They show that XY rabbits present with complete XY male-to-female sex reversal, very similar to what observed in human 46, XY DSD patients (but not the mice model). They further show that in the XY sex reversed gonads, germ cells fail to enter meiosis. They next analysed XX gonads and while there is no major effect on sex determination (as expected), the germ cells in these ovaries fail to enter meiosis, highlighting the critical role that DMRT1 has in germ cells.

I think it is really important that we start to embrace other mammal models that are not the mouse as we find many instances that the mouse is not the optimal system for understanding human sex determination.

The study is well explained and presented. The data is clear, and the paper is fluent to read.

Reviewer #3 (Public Review):

This manuscript deal with the sex-related gene, DMRT1, showing that it has a testis-promoting function in the rabbit. Loss-of-function studies the mouse and human, DMRT1 has a role in testis maintenance after birth, although forced expression in mouse can induce testis formation.

The authors used CRISPR/Cas9 genome editing to generate DMRT1^{-/-} rabbit embryos. The gonads of these embryos developed as ovaries. Interestingly, in addition Y-linked SRY, DMRT1 is required for timely up-regulation of SOX9 during Sertoli cell differentiation in the male gonad. This is quite different to the situation in mouse, where *Dmrt1* is not required in the testis until after birth (and Sry induced up-regulation of Sox9 hence does not require *Dmrt1*).

The work adds to the field of sex determination by further broadening our understanding of the DMRT1 gene and the evolution of gonadal sex determination.

In the Discussion section, it is suggested that DMRT1 could act as a pioneering factor to allow SRY action upon Sox9 in the rabbit model. The data show that DMRT1 may be more central to testis formation in mammals than previously considered. The work supports the notion that our understanding that the genetics of gonadal development (and indeed development more generally) should not rest solely on findings in the mouse.

Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Recommendations For The Authors):

Some sentences need to be clarified and some additional data and references could be added.

1. Line 18

SRY is the sex-determining gene

Modification done

1. Line 50

Despite losing its function in early testis determination in mice, DMRT1 retained part of this function in adulthood when it is necessary to maintain Sertoli cell identity.

Losing its function is misleading. The authors describe firstly that Dmrt1 has no obvious function in embryonic testis development but is critical for the maintenance of Sertoli cells in adult mice. The wording "losing its function in early testis" is confusing. Do the authors mean that despite the expression of Dmrt1 in early testis development, the function of Dmrt1 seems to be restricted to adults in mice? A comparison between the testis and ovary should be more cautious since GarciaAlonso et al (2022) have shown that the transcriptomics of supporting cells between humans and mice is partly different.

That's what we thought, and the sentence has been changed as follow: "Although DMRT1 is not required for testis determination in mice, it retained part of its function in adulthood when it is necessary to maintain Sertoli cell identity." (line 51 to 53)

1. Line 78

XY DMRT1^{-/-} rabbits showed early male-to-female sex reversal.

Sex reversal indicates that there is no transient Sertoli cell differentiation that transdifferentiate into granulosa cells. This brings us to an interesting point. In the case of reprogramming, the transient Sertoli cells can produce AMH leading to the regression of the Mullerian ducts. In humans, some 9pdeleted XY patients have Mullerian duct remnants and feminized external genitalia. This finding indicates early defects in testis development.

Is there also feminized external genitalia in XY Dmrt1^{-/-} rabbits. Can the authors comment on the phenotype of the ducts?

We proposed to add "and complete female genitalia" at the end of the following sentence: "Secondly, thanks to our CRISPR/Cas9 genetically modified rabbit model, we demonstrated that DMRT1 was required for testis differentiation since XY DMRT1^{-/-} rabbits showed early male-tofemale sex reversal with differentiating ovaries and complete female genitalia." (line 77 to 80)

Indeed, since the first stage (16 dpc) where we can predict the sex of the individual by observing its gonads during dissection, we always predict a female sex for XY DMRT1 KO fetuses. It is only genotyping that reveals an XY genotype. At birth, our rabbits are sexed by technicians from the facility and again, but now based on the external genitalia, they always phenotype these rabbits as female ones. In these XY KO rabbits, the supporting cells never differentiate into Sertoli, and ovarian differentiation occurs as early as in XX animals. Thus, these animals are fully feminized with female internal and external genitalia. Most of 9p-deleted patients are not homozygous for the loss-offunction of DMRT1, and the remaining wild-type allele could explain the discrepancy between KO rabbits and humans.

1. Line 53

In the ovary, an equivalent to DMRT1 was observed since FOXL2 (Forkhead family box L2) is expressed in female supporting cells very early in development.

Can the authors clarify what is the equivalent of DMRT1, is it FOXL2? DMRT1 heterozygous mutations result in XY gonad dysgenesis suggesting haploinsufficiency of DMRT1. However, to my knowledge, there is no evidence of haploinsufficiency in XX babies. Thus can we compare testis and ovarian genetics?

We agree, the term “equivalent” is ambiguous, and we changed the sentence as follows: “In ovarian differentiation, FOXL2 (Forkhead family box L2) showed a similar function discrepancy between mice and goats as DMRT1 in the testis pathway. In the mouse, Foxl2 is expressed in female supporting cells early in development but does not appear necessary for fetal ovary differentiation. On the contrary, it is required in adult granulosa cells to maintain female-supporting cell identity.” (line 53 to 56)

Regarding reviewer 2's question on haploinsufficiency in humans: the patient described in Murphy et al., 2015 is an XY individual with complete gonadal dysgenesis. But, it has been shown that the mutation carried by this patient leads to a dominant-negative protein, equivalent to a homozygous state (Murphy et al., 2022).

For FOXL2 mutation in XX females, haploinsufficiency does not affect early ovarian differentiation (no sex reversal) but induces premature ovarian failure.

We agree with the reviewer, we cannot compare testis and ovarian genetics considering two different genes.

1. Line 55

In mice, Foxl2 does not appear necessary for fetal ovary differentiation (Uda et al., 2004), while it is required in adult granulosa cells to maintain female-supporting cell identity (Ottolenghi et al., 2005). The reference Uhlenhaut et al (2009) reporting the phenotype of the deletion of Foxl2 in adults should be added.

The reference has been added.

1. Line 64

These observations in the goat suggested that DMRT1 could retain function in SOX9 activation and, thus, in testis determination in several mammals.

Lindeman et al (2021) have shown that DMRT1 can act as a pioneer factor to open chromatin upstream and Dmrt1 is expressed before Sry in mice (Raymond et al, 1999, Lei, Hornbaker et al, 2007). Whereas additional factors may compensate for the absence of Dmrt1, these results suggest that DMRT1 is also involved in Sox9 activation.

Dmrt1 is indeed expressed before Sry/Sox9 in the mouse gonad. However, no binding site for DMRT1 could be observed at Sox9 enhancer 13 in mice. This does not support a role for DMRT1 in the activation of Sox9 expression in this species. Furthermore, in Lindeman et al 2021, the authors clearly state that DMRT1 acts as a pioneering factor for SOX9 only after birth. It does not appear to have this role before. One of the explanations put forward is that the state of chromatin is different during fetal development in mice: chromatin is more permissive and does not require a factor to facilitate its opening. This hypothesis is based in particular on the description of a similar chromatin profile in the precursors of XX and XY fetal supporting cells, where many common regions display an open structure (Garcia-Moreno et al., 2019). Once sex determination and differentiation are established, a sex-specific epigenome is set up in gonadal cells. Chromatin remodeling agents are then needed to regulate gene expression. We hypothesize that in non-murine mammals such as rabbits, the state of gonadal cell chromatin would be different in the fetal period, more repressed, requiring the intervention of specific factors for its opening, such as DMRT1.

1. Figure 1

Most of the readers might not be familiar with the developmental stages of the gonad in rabbits. A diagram of the key stages in gonad development would facilitate the understanding of the results.

Thank you, it has been added in Figure 1.

1. Figure 2

Arrowheads are difficult to spot, could the authors use another color?

Done

1. Line 117: can the authors comment on the formation of the tunica albuginea? Do the epithelial cells acquire some specific characteristics?

The formation of the tunica albuginea begins with the formation of loose connective tissue beneath the surface epithelium of the male gonad. The appearance of this tissue is concomitant with the loss of expression of DMRT1 in the cell of the coelomic epithelium. Our interpretation is that the contribution of the cells from the coelomic epithelium and their proliferation stops when the tunica begins to form because the structure of the tissue beneath the epithelium change, and the cellular interactions between the epithelium and the tissue below remain disrupted. By contrast, these interactions persist in the ovary until around birth for ovigerous nest formation.

1. The first part of the results described DMRT1 expression in rabbits. With the new single-cell transcriptomic atlas of human gonads, it would be important to describe the pattern of expression in this species. This could be described in the introduction in order to know the DMRT1 expression pattern in the human gonad before that of the rabbit.

A comment on the expression pattern of DMRT1 in human fetal gonads has been added in the discussion section: "In the human fetal testis, DMRT1 expression is co-detected with SRY in early supporting gonadal cells (ESCGs), which become Sertoli cells following the activation of SOX9 expression (Garcia-Alonso et al., 2022) » (line 222 to 224)

1. Figure 3 supplement 3

Dotted line: delimitation of the ovarian surface epithelium. Could the authors check that there is a dotted line?

Done

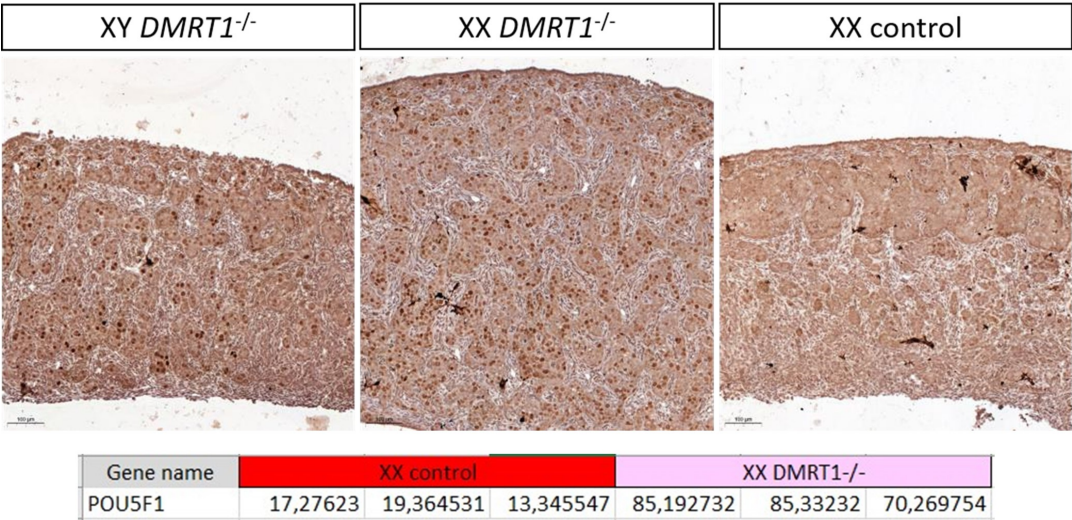
1. Figure 5 and Line 186

Quantification is missing such as the % of germ cells, % of meiotic germ cells.

Quantification is not easy to realize in rabbits because of the size and the elongated shape of the gonad. Indeed, it's difficult to be sure that both sections (one from WT, the other from KO) are strictly in a similar region of the gonad and that the section is perfectly longitudinal or not. See also our answer to reviewer 3 (point 7) on this aspect. Actually, we are trying to make a better characterization of this XX phenotype and to find a marker of the pre-leptotene/leptotene stage susceptible to work in rabbits (SYCP3 will be the best, but we encountered huge difficulties with different antibodies and even RNAscope probe!). So

actually, the most convincing indirect evidence of this pre-meiotic blockage (in addition to HE staining at 18 dpp in the new Figure 6) is the persistence of POU5F1 (pluripotency), specifically in the germinal lineage of KO XX and XY gonads. In addition to the new figure supplement 5, we can show you in Author response image 1: (i) the gonadal section at a lower magnification, where it is evident that there is a big difference between WT and KO germ cell POU5F1-stainings; and (ii) POU5F1 expression from a bulk RNA-seq realized the day after birth at 1 dpp where the difference is also transcriptionally very clear.

Author response image 1.



1. Line 186,
E is missing at preleptoten

Added

1. Figure supplement 7.
A magnification of the histology of the gonads is missing.

This figure is only for showing the gonadal size, and there are the same gonads as in the new Figure 6. So, the magnification is represented in Figure 6.

15)Discussion
Line 201
SOX9, well known in vertebrates,

The references of the human DSD associated with *SOX9* mutations are missing. Thank you, references have been added.

1. Line 286
One of the targets of WNT signaling is Bmp2 in the somatic cells and in turn, Zglp1, which is required for meiosis entry in the ovary as shown by Miyauchi et al (2017) and Nagaoka et al (2020). Does the level of BMP pathway vary in DMRT1 mutants?

At 20 dpc, the expression level of BMP2 in XY and XX DMRT1 mutants gonads is similar to the one of XX control which is lower than in XY control (see the TMP values from our RNA-seq in Author response image 2).

Author response image 2.

Gene name	XY Control			XY DMRT1 ^{-/-}			XX DMRT1 ^{-/-}			XX Control		
BMP2	3,74766405	4,47640596	3,19672684	0,58310189	3,28465296	0,97740781	1,12736639	1,22681909	1,0260926	1,23558509	1,33650813	0,77656961
BMP4	17,6598037	16,7636931	11,3165933	4,64798232	14,2750166	4,24295372	4,48799752	6,4980386	9,44063234	9,38648951	6,76878759	5,22522002
ZGLP1	6,64611768	6,64857137	5,33248662	7,16285747	5,19478918	7,89000816	7,77256289	4,1191315	4,65028165	5,85335302	7,76907021	8,82924755

Reviewer #2 (Recommendations For The Authors):

Here are my minor comments:

1. Line 106- You mention that coelomic epithelial cells only express DMRT1. Please add an arrow to highlight where you refer to.

Done

1. Line 112: In mice, the SLCs also express Sox9 but not Sry apart from Pax8. You mention here that the SLCs are expressing SRY and DMRT1 in addition to PAX8. Could you perhaps explain the difference? Please refer to that in the results or discussion.

We add a new sentence at the end of this paragraph on SLCs: “As in mice, these cells will express SOX9 at the latter stages (few of them are already SOX9 positive at 15 dpc), but unlike mice, they express SRY.” (line 114 to 115)

We already have collaborations with different labs on these SLC cells, and we will certainly come back later on this aspect, remaining slightly off-topic here.

1. Could you please explain why did you chose to target Exon 3 of DMRT1 and not exons 1-2 which contain the DM domain? Was it to prevent damaging other DMRT proteins? Is there an important domain or function in Exon 2?

Our choice was mainly based on technical issues (rabbit genome annotation & sgRNA design), but also we want to avoid targeting the DM domain due to its strong conservation with other DMRT genes. Due to the poor quality of the rabbit genome, exons 1 and 2 are not well annotated in this species. We have amplified and sequenced the region encompassing exons 1 & 2 from our rabbit line, but the software used for sgRNA design does not predict good guides on this region. The two best sgRNAs were predicted on exon 3, and we used both to obtain more mutated alleles.

1. Your scheme in Supp Figure 4 is not so clear. It is not clear that the black box between the two guides is part of Exon 3 (labelled in blue).

The scheme has been improved.

1. Did you only have 1 good founder rabbit in your experiment? Why did you choose to work with a line that had duplication rather than deletion?

Very good point! In the first version of this paper, we'd try to explain the long (around 2 years) story of breeding to obtain the founder animal. Here it is:

During the genome editing process, we generate 6 mosaic founder animals (5 males and 1 female), then we cross them with wild-type animals to isolate each mutated allele in F1 offspring used afterward to establish and amplify knockout lines. Unexpectedly, we observe a very slow ratio of mutated allele transmission (5 on 129 F1 animals), and only one mutated allele has been conserved from the unique surviving adult F1 animal. It consists of an insertion of the deleted 47 bp DNA fragment, flanked by the cutting sites of the two RNA guides used with Cas9.

The main hypothesis to explain this mutation event is that in the same embryonic cell, the deletion occurs on one allele then the deleted fragment remains inserted into the other allele. Under this scheme, the embryonic cell carries a homozygous DMRT1 knockout genotype, albeit heterogeneous, with a deleted allele (del47) and the present allele (insertion of a 47 bp fragment leading to an in sense duplication). This may explain the very low frequency of transmission since all germ cells carrying a homozygous DMRT1^{-/-} genotype will probably not be able to enter the meiotic process as suggested by our results on XX and XY DMRT1^{-/-} ovaries. Finally, and under this hypothesis, the way we obtained this unique founder animal remains a mystery!

1. Figure 4- real-time data- where does it say what is a,b,c,d of the significance? It should appear on the figure itself and not elsewhere.

Modification done.

1. If I understand correctly, you were able to get the rabbits born and kept to adulthood (you show in supp figure 7 their gonads). What was the external phenotype of these rabbits? Did the XY mutant gonads have the internal and external genitals of a female (oviduct, uterus, vagina etc.)?

See our answer to Reviewer 1 on this question (point 3).

1. Line 20: It is more correct to write 46, XY DSD rather than XY DSD

Modification done.

1. Line 21: you can remove the "the" after abolished

Modification done.

1. Line 31: consider replacing the first "and" by "as well as" since the sentence sounds strange with two "and".

Modification done.

1. Line 212- Please check with the eLife guidelines if they allow "data not shown" in the paper.

This is unspecified.

Reviewer #3 (Recommendations For The Authors):

The following points should be addressed.

1. The *in situ*'s in Fig 1 and 2 are very clear. Fig 1 and Fig 2, *In situ* hybridisation in tissue sections, it looked like DMRT1 could be expressed in some cells where SRY mRNA is

absent @ E13.5dpc and 14.5 dpc. Do you think this is real, or maybe Sry is turned off now in those cells?

Based on the results of in situ hybridizations, DMRT1 appears to be expressed by both coelomic epithelium and genital crest medullar cells in a pattern that is actually broader than that of SRY. Moreover, in rabbits, SRY expression seems to start in the medulla of the genital ridge rather than in the surface epithelium, as described in mice (see Figure 1 at 12 and 13 dpc). Nevertheless, more detailed analyses are needed to ensure the lineage of cells expressing SRY and/or DMRT1, such as single-cell RNAseq at these key stages of sexual determination in rabbits (from 12 to 16 dpc).

1. It is curious that SRY expression is elevated in the DMRT1 KO (Knockout) rabbit gonads. Does this suggest feedback inhibition by DMRT1, or maybe indirect via effect on Sox9 (as I believe Sox9 feeds back to down-regulate Sry in mouse, for example).

The maintenance of SRY expression in the DMRT1 $-/-$ rabbit testis seems to be linked to the absence of SOX9 expression. We believe that, as in mice, SOX9 would down-regulate SRY (even if, in rabbits, SRY expression is never completely turned off).

1. I suggest the targeting strategy and proof of DMRT1 knockout by sequencing etc. be brought out of the suppl. Data and shown as a figure in the text.

See also our answer to reviewer 2 (point 5). It has needed huge efforts to obtain these DMRT1 mutated rabbit line, and of course, it constitutes the basis of the study. But regarding the title and the main message of the article, we are not convinced that the targeting strategy should be moved into the main text.

1. Unless there are limitations imposed by the journal, I also feel that Suppl Fig 5 (the immunostaining) deserves to be in the paper text too. The Fig showing loss of DMRT1 by immunostaining is important.

We include the figure supplement 5 in the main text. So, Figure 4E and figure supplement 5 have been combined into a new Figure 5.

1. The RT-qPCR data should have the statistics clarified on the graphs. (e.g., it is stated that, although Sox9 mRNA is clearly down, there is a slight increase compared to control on KO XX gonads. Is this statistically significant? Figure legend states that the Kruskal-Wallis test is used, and significance is shown by letters. This is unclear. It would be better to use the more usual asterisks and lines to show comparisons.

Modification done.

1. Reference is made to DMRT1 $\pm$ rabbits having aberrant germ cell development, pointing to a dosage effect. This is interesting. Does the somatic part of the gonad look completely normal in the het knockouts?

DMRT1 heterozygous male rabbits have a phenotype of secondary infertility with aging, and we are trying now to better characterize this phenotype. The problem is complex because, as we cannot carry out conditional KO, it remains difficult to decipher the consequence of DMRT1 haploinsufficiency in the Sertoli cells versus the germinal ones. Anyway, the somatic part is sufficiently normal to support spermatogenesis since heterozygous males are fertile at puberty and for some months thereafter.

1. *Can the authors indicate why meiotic markers were not used to explore the germ cell phenotype? It would be advantageous to use a meiotic germ cell marker to definitely show that the germ cells do not enter meiosis after DMRT1 loss. (Not just H/E staining or maintenance of POU). Example SYCP3, or STRA8 (as pre-meiotic marker) by in situ or immunostaining. Even though no germ cells were detected in adult KO gonads.*

The expression of pre-meiotic or meiotic markers is currently under study in DMRT1^{-/-} females. Transcriptomic data (RNA-seq) are also being analyzed. We are preparing a specific article on the role of DMRT1 in ovarian differentiation in rabbits. We felt it was important to reveal the phenotype observed in females in this first article, but we still need time to refine our description and understanding of the role of DMRT1 in the female.

1. *What future studies could be conducted? In the Discussion section, it is suggested that DMRT1 could act as a pioneering factor to allow SRY action upon Sox9. How could this be further explored?*

To explore the function of DMRT1 as a pioneering factor, it now seems necessary to characterize the epigenetic landscapes of rabbit fetal gonads expressing or not DMRT1 (comparison of control and DMRT1^{-/-} gonads). Two complementary approaches could be privileged: the study of chromatin opening (ATAC-seq) and the analysis of the activation state of regulatory regions (CUT&Tag). The study of several histone marks, such as H3K4me3 (active promoters), H3K4me1 (primed enhancers), H3K27ac (enhancers and active promoters), and H3K27me3 (enhancers and repressed promoters), would be of great interest. However, these techniques are only relevant for gonads that can be separated from the adjacent mesonephros, which is only possible from the 16 dpc stage in rabbits. To perform a relevant analysis at earlier stages, a "single-nucleus" approach such as ATAC-seq single-nucleus or multi-omic single-nucleus combining ATAC-seq and RNA-seq could be used.